

Supplementary Material: How Much History Is Enough? Certifying Decision-Relevant Memory in Offline Reinforcement Learning

Anonymous submission

Notation

At stage h , X_h is the full observable history and $\phi_{m,h}(X_h)$ is its m -observation suffix. The suffix cell is $C_{m,h}(z) = \{x : \phi_{m,h}(x) = z\}$. We use $A_h^*(x, a) = V_h^*(x) - Q_h^*(x, a)$,

$$\eta_{m,h}(z) = \min_{p \in \Delta(\mathcal{A}_h)} \max_{x \in C_{m,h}(z)} \mathbb{E}_{a \sim p} A_h^*(x, a),$$

$\eta_{m,h} = \max_z \eta_{m,h}(z)$, and $G_m = \sum_h \eta_{m,h}$.

Population Proofs

Lemma 1 (Monotonicity). *For every stage, $\eta_{m+1,h} \leq \eta_{m,h}$, and therefore $G_{m+1} \leq G_m$.*

Proof. Every $(m+1)$ -suffix cell C' lies inside one m -suffix cell C . Let p_C minimize the objective for C . Then

$$\begin{aligned} \eta(C') &\leq \max_{x \in C'} \mathbb{E}_{a \sim p_C} A_h^*(x, a) \\ &\leq \max_{x \in C} \mathbb{E}_{a \sim p_C} A_h^*(x, a) \\ &= \eta(C). \end{aligned}$$

Taking the maximum over refined cells proves the stagewise claim; summing proves the cumulative claim. $\square$

Theorem 1 (Exact sufficiency). *For a fixed m , $G_m = 0$ if and only if there exists an m -suffix policy that is optimal from every full context at every stage.*

Proof. All entries of A_h^* are nonnegative. Thus a cell has zero minimax value if and only if some action distribution $p_{h,z}$ has zero expected advantage loss at every context in the cell. Use these distributions to form a suffix policy π . At stage H , $V_H^\pi(x) = \mathbb{E}_{a \sim p_{H,z}} Q_H^*(x, a) = V_H^*(x)$. If the equality holds at stage $h+1$, then

$$\begin{aligned} V_h^\pi(x) &= \mathbb{E}_{a \sim p_{h,z}} r_h(x, a) \\ &\quad + \mathbb{E}_{a \sim p_{h,z}} P_h V_{h+1}^\pi(x, a) \\ &= \mathbb{E}_{a \sim p_{h,z}} Q_h^*(x, a) \\ &= V_h^*(x). \end{aligned}$$

This proves sufficiency by backward induction.

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Conversely, suppose π is uniformly optimal. Backward induction from the terminal stage shows that its continuation after every action is also valued by V_{h+1}^* , hence

$$V_h^*(x) = V_h^\pi(x) = \mathbb{E}_{a \sim \pi_h(z)} Q_h^*(x, a).$$

The expected optimal advantage loss is zero in every context of every cell, so $G_m = 0$. $\square$

Theorem 2 (Value-loss sandwich). *Let $\mathcal{E}_m$ denote the best uniform loss attainable by Π_m :*

$$\mathcal{E}_m = \inf_{\pi \in \Pi_m} \max_{h,x} [V_h^*(x) - V_h^\pi(x)].$$

Then

$$\max_h \eta_{m,h} \leq \mathcal{E}_m \leq \sum_h \eta_{m,h}.$$

Proof. For the upper bound, let π^m use a minimax action distribution in each cell and define $D_h(x) = V_h^*(x) - V_h^{\pi^m}(x)$. Since $Q_h^*(x, a) = r_h(x, a) + P_h V_{h+1}^*(x, a)$,

$$\begin{aligned} D_h(x) &= \mathbb{E}_{a \sim \pi_h^m(z)} A_h^*(x, a) \\ &\quad + \mathbb{E}_{a, X'} D_{h+1}(X') \\ &\leq \eta_{m,h} + \|D_{h+1}\|_\infty. \end{aligned}$$

With $D_{H+1} = 0$, backward induction yields $D_h(x) \leq \sum_{t=h}^H \eta_{m,t}$.

For the lower bound, fix any $\pi \in \Pi_m$. Since $V_{h+1}^\pi \leq V_{h+1}^*$,

$$V_h^\pi(x) \leq \mathbb{E}_{a \sim \pi_h(z)} Q_h^*(x, a),$$

and therefore

$$V_h^*(x) - V_h^\pi(x) \geq \mathbb{E}_{a \sim \pi_h(z)} A_h^*(x, a).$$

For each stage, maximize over contexts within cells and then over cells. Any shared action distribution incurs at least $\eta_{m,h}$, so the uniform loss is at least $\max_h \eta_{m,h}$. Taking the infimum over π completes the proof. $\square$

Observation Prediction versus Decision Sufficiency

Fix a behavior policy μ and define

$$\begin{aligned} R_m^{\text{obs}} &= \sum_h H_\mu(O_{h+1} \mid \phi_{m,h}(X_h), A_h) \\ &\quad - \sum_h H_\mu(O_{h+1} \mid X_h, A_h). \end{aligned}$$

In the delayed-cue construction, all observations after the initial cue are constant, so $R_m^{\text{obs}} = 0$ for every m , whereas the final action incurs minimax advantage loss $\gamma/2$ until the cue enters the suffix. In the nuisance construction, the initial bit reappears as the terminal observation but action zero is optimal after either bit. Hence $G_m = 0$ for every m , while $R_m^{\text{obs}} = 1$ bit for all $m < H$ under a balanced cue.

This incomparability concerns observation-only prediction. The exact controlled Markov condition is stronger. Suppose that for each h, m, z, a , the conditional joint law of $(R_h, \phi_{m,h+1}(X_{h+1}))$ is identical for all $x \in C_{m,h}(z)$. Backward induction then shows that $Q_h^*(x, a)$ is constant within every suffix cell for each action. A common maximizing action therefore exists, and $G_m = 0$. Thus full reward–transition predictive sufficiency upper-bounds decision-relevant memory; the potentially two-sided failures arise for observation-prediction diagnostics.

Offline Certificate Proofs

Assume simultaneous intervals $L_h \leq Q_h^* \leq U_h$ and define

$$\bar{A}_h(x, a) = \max\{0, \max_{b \neq a} [U_h(x, b) - L_h(x, a)]\}.$$

The self-comparison is omitted because it is exactly zero.

Lemma 2 (Upper advantage). *On the interval event, $A_h^*(x, a) \leq \bar{A}_h(x, a)$ for all h, x, a .*

Proof. For every $b \neq a$, interval validity gives

$$Q_h^*(x, b) - Q_h^*(x, a) \leq U_h(x, b) - L_h(x, a).$$

Taking the positive maximum over competitors proves $A_h^*(x, a) \leq \bar{A}_h(x, a)$. $\square$

Theorem 3 (Simultaneous value certificate). *Let $\bar{\pi}^m$ be the suffix policy obtained by replacing A_h^* with $\bar{A}_h$ in each cellwise minimax problem. Then, on the interval event, simultaneously for all m, h, x ,*

$$V_h^*(x) - V_h^{\bar{\pi}^m}(x) \leq \sum_{t=h}^H \bar{\eta}_{m,t}.$$

Proof. The proof is the upper-bound recursion above, with

$$\mathbb{E}_{a \sim \bar{\pi}_h^m(z)} A_h^*(x, a) \leq \mathbb{E}_{a \sim \bar{\pi}_h^m(z)} \bar{A}_h(x, a) \leq \bar{\eta}_{m,h}.$$

The interval event is simultaneous across states, actions, stages, and memory lengths, so a data-dependent choice among the candidate memories remains covered. $\square$

Proposition 1 (Interval tightness). *If every one-sided interval error is at most w , then*

$$A_h^* \leq \bar{A}_h \leq A_h^* + 2w, \quad \eta_{m,h} \leq \bar{\eta}_{m,h} \leq \eta_{m,h} + 2w,$$

and $G_m \leq \bar{G}_m \leq G_m + 2Hw$.

Proof. For every competitor $b \neq a$,

$$U_h(x, b) - L_h(x, a) \leq Q_h^*(x, b) - Q_h^*(x, a) + 2w.$$

Taking the positive maximum gives the pointwise cost inequality. The lower LP inequality follows from pointwise domination. For the upper LP inequality, evaluate the robust objective at an exact minimizer; adding at most $2w$ to each action cost adds at most $2w$ to every mixture cost. Sum over stages. $\square$

Corollary 1 (Exact memory recovery). *If $m_{\text{dec}} = \min\{m : G_m = 0\} > 1$, let $\Gamma = \min_{m < m_{\text{dec}}} G_m$. If $2Hw < \Gamma$, then every threshold $\epsilon \in [2Hw, \Gamma)$ makes DRMS select m_{dec} on the interval event.*

Proof. Every shorter memory has $\bar{G}_m \geq G_m \geq \Gamma > \epsilon$, whereas $\bar{G}_{m_{\text{dec}}} \leq 2Hw \leq \epsilon$. $\square$

Tabular Confidence Intervals

Let $M = \sum_h |\mathcal{X}_h| |\mathcal{A}_h|$. For visit count $N_h(x, a)$, use reward and transition radii

$$\beta_h^r = \sqrt{\frac{\log(4M/\delta)}{2 \max\{1, N_h\}}},$$

with radius one if $N_h = 0$, and

$$\beta_h^P = \min \left\{ 2, \sqrt{\frac{2(|\mathcal{X}_{h+1}| \log 2 + \log(2M/\delta))}{\max\{1, N_h\}}} \right\}.$$

Hoeffding’s inequality, the L_1 deviation inequality of Weissman et al. (2003), and a union bound give simultaneous model coverage.

Initialize $\underline{V}_{H+1} = \bar{V}_{H+1} = 0$ and recurse

$$\begin{aligned} \underline{Q}_h &= \text{clip} \left(\hat{r}_h - \beta_h^r + \hat{P}_h \underline{V}_{h+1} - \frac{1}{2} \beta_h^P \text{span}(\underline{V}_{h+1}) \right), \\ \bar{Q}_h &= \text{clip} \left(\hat{r}_h + \beta_h^r + \hat{P}_h \bar{V}_{h+1} + \frac{1}{2} \beta_h^P \text{span}(\bar{V}_{h+1}) \right), \end{aligned}$$

where clipping is to $[0, H - h + 1]$ and each value bound is the actionwise maximum of its Q -bound.

Proposition 2 (Validity). *On the simultaneous model event, $\underline{Q}_h \leq Q_h^* \leq \bar{Q}_h$ at every state-action-stage triple.*

Proof. Proceed backward. If $\underline{V}_{h+1} \leq V_{h+1}^* \leq \bar{V}_{h+1}$, then

$$\begin{aligned} Q_h^* &= r_h + P_h V_{h+1}^* \\ &\geq \hat{r}_h - \beta_h^r + P_h \underline{V}_{h+1} \\ &\geq \hat{r}_h - \beta_h^r + \hat{P}_h \underline{V}_{h+1} - \frac{1}{2} \beta_h^P \text{span}(\underline{V}_{h+1}). \end{aligned}$$

The upper inequality is symmetric. Maximization preserves value bounds, and clipping preserves the bounds because remaining returns lie in $[0, H - h + 1]$. $\square$

Matching Rates in the Two-Point Family

Let $B \sim \text{Bernoulli}(p)$ be an observed stage-one cue and let the stage-two current observation be constant. The behavior policy chooses the two final actions uniformly. Write

$$r_+ = \frac{1}{2} + \frac{\gamma}{2}, \quad r_- = \frac{1}{2} - \frac{\gamma}{2}.$$

In model $\mathcal{M}_0$, action zero has Bernoulli mean r_+ after either cue and action one has mean r_- . In model $\mathcal{M}_1$, these means are swapped only after $B = 1$. Therefore $m_{\text{dec}}(\mathcal{M}_0) = 1$ and $m_{\text{dec}}(\mathcal{M}_1) = 2$.

Theorem 4 (Lower bound). *For $0 < \gamma \leq 1/2$ and $0 < \delta < 1/4$, any estimator with error at most δ in both models requires*

$$n \geq \frac{\log(1/(4\delta))}{6p\gamma^2}.$$

Proof. The models differ only after the rare cue. Under the uniform final action, the one-episode divergence is

$$D_{\text{KL}}(P_0 \| P_1) = \frac{p}{2} [\text{kl}(r_+ \| r_-) + \text{kl}(r_- \| r_+)].$$

For Bernoulli variables, $\text{kl}(u \| v) \leq (u - v)^2 / [v(1 - v)]$. Since $r_- r_+ = 1/4 - \gamma^2/4 \geq 3/16$, each KL term is at most $(16/3)\gamma^2$. Thus $D_{\text{KL}}(P_0 \| P_1) \leq (16/3)p\gamma^2 < 6p\gamma^2$.

The Bretagnolle–Huber inequality gives, for any test,

$$\mathbb{P}_0(\text{error}) + \mathbb{P}_1(\text{error}) \geq \frac{1}{2} \exp[-D_{\text{KL}}(P_0^n \| P_1^n)].$$

If both errors are at most δ , then $2\delta \geq \frac{1}{2}e^{-6np\gamma^2}$. Rearrangement proves the result. $\square$

Corollary 2 (Tabular DRMS upper bound). *Assume $p \leq 1/2$, and let M be the total number of full-history state–action pairs. Run the tabular interval construction with confidence parameter $\delta/2$, and run DRMS with tolerance $\gamma/4$. If*

$$n \geq \frac{32 \log(8M/\delta)}{p\gamma^2},$$

then DRMS returns the correct memory in both models with probability at least $1 - \delta$.

Proof. Every final state–action count has binomial mean at least $np/2$. A multiplicative Chernoff bound and a union bound over the four final pairs give

$$\mathbb{P}\left(\min_{x,a} N_2(x, a) < np/4\right) \leq 4e^{-np/16}.$$

The stated sample size, together with $\gamma \leq 1/2$, makes this at most $\delta/2$. On the simultaneous reward-concentration event, which has probability at least $1 - \delta/2$, every final reward radius obeys

$$\beta_2^r(x, a) \leq \sqrt{\frac{2 \log(8M/\delta)}{np}} \leq \frac{\gamma}{4}.$$

Because the empirical mean can deviate from the truth by at most the same radius, $L_2(x, a_*) \geq Q_2^*(x, a_*) - 2\beta_2^r$ and $U_2(x, a) \leq Q_2^*(x, a) + 2\beta_2^r$. Hence the true best action in each final context has robust upper advantage zero. In $\mathcal{M}_0$, action zero is common to both contexts, so $\bar{G}_1 = 0$. In $\mathcal{M}_1$, $\bar{G}_1 \geq G_1 = \gamma/2 > \gamma/4$, while each full-memory cell has $\bar{G}_2 = 0$. DRMS therefore selects one and two, respectively. $\square$

Additional Experiment

All tests, raw outputs, and scripts are provided in the accompanying repository.

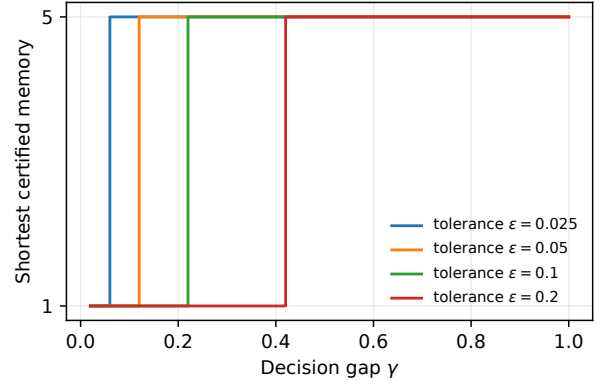


Figure 1: Population tolerance phase transition at horizon five. An insufficient suffix has ambiguity $\gamma/2$, so memory one is certified exactly when $\gamma \leq 2\epsilon$.

References

Weissman, T.; Ordentlich, E.; Seroussi, G.; Verdú, S.; and Weinberger, M. J. 2003. Inequalities for the L1 Deviation of the Empirical Distribution. Technical Report HPL-2003-97R1, Hewlett-Packard Laboratories.